output :

Survived	Pclass		Sex	age	sibsp	Parch	fare	embarked	class
0	0	3	male	22.0	1	0	7.2500	S	Third
1	1	1	female	38.0	1	0	71.2833	C	First
2	1	3	female	26.0	0	0	7.9250	S	Third
3	1	1	female	35.0	1	0	53.1000	S	First
4	0	3	male	35.0	0	0	8.0500	S	Third

Dataset info and missing values Summary

Dataset info :

<class 'pandas. core. frame. DataFrame'>

Range Index: 891 entries, 0 to 890

Data columns (total 15 columns):

Missing values

Survived	0	embarked	2
Pclass	0	class	0
sex	0	who	0
age	177	adult_male	0
sibsp	0	deck	688
parch	0	embark_town	2
fare	0	alive	0

EX. NO - 01
22/7/25

Load the Titanic dataset & convert it into a Data Frame

Aim:

To perform basic Preprocessing and exploratory data analysis (EDA) on titanic dataset using Pandas, seaborn and Sklearn tools.

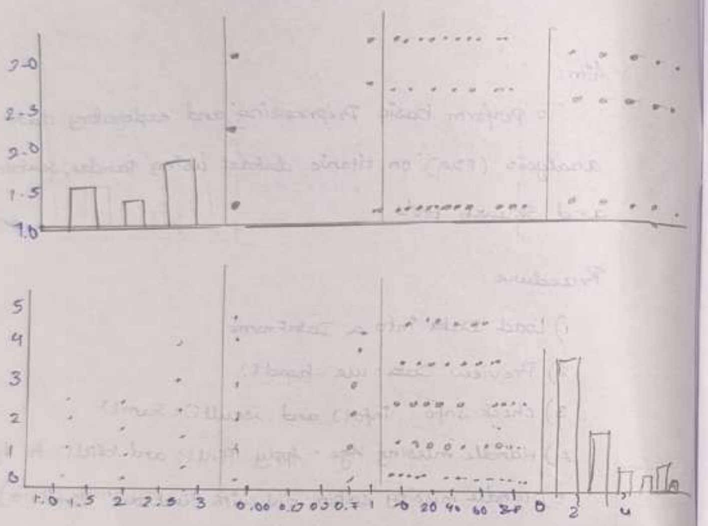
Procedure:

1) Load Data into a DataFrame.
2) Preview Data: use .head()
3) check Info: . info() and .isnull().Sum()
4) Handle missing Age: Apply . ffill() and bfill() to 'Age'
5) Handle missing cabin: Fill with "unknown" (limits 5)
6) Remove Duplicates: use . drop_duplicates()
7) Encode 'Sex': use LabelEncoder.
8) Scale 'Fare': use standard Scaler
9) Pairplot: For 'pclass', 'sex', 'Age', 'sibsp'.
10) Heatmap: For 'pclass', 'Age', 'sibsp', 'parch', 'Fare'.

Program:

```
import Pandas as pd
import seaborn as sns
Import matplotlib.pyplot as plt
from sklearn. Preprocessing import label Encoder, standard Scaler
df = sns.load_dataset ('titanic')
Print ("First 5 rows:")
Print(df.head ())
Print ("\n Dataset info:")
df info()
Print ("\n Missing values : \n", df.isnull().sum())
df['Age_ffill'] = df['age'].ffill()
df['Age_bfill'] = df['age'].bfill()
```

Pair plot



Fare parch sibsp pclass

Pclass	age	sibsp	Parch	fare
1.00	-0.37	0.09	0.02	-0.55
-0.37	1.00	-0.32	-0.20	0.09
0.09	-0.32	1.00	0.41	0.15
0.02	-0.30	0.41	1.00	0.21
-0.13	0.09	0.15	0.21	1.00

```
df['deck'] = df['deck'].cat.add_categories ('unknown')
df['deck'] = df['deck'].fillna('unknown', limit=5)
df.drop_duplicates(inplace = True)
le = Label Encoder()
df['Sex_encoded'] = le.fit = transform (df['Sex'])
Scaler = standard_Scaler()
df['fare_scaled'] = Scaler.fit-transform(df[['fare']])
Selected_features = ['pclass', 'Sex_encoded', 'age', 'sibsp']
sns.pairplot (df[selected_features].dropna())
plt.suptitle ("Pair plot", y = 1.02)
plt.show()
Corr_features = ['pclass', 'age', 'sibsp', 'Parch', 'fare']
corr-matrix = df[corr-features]. corr()
plt.figure (fig size = (8,6))
sns.heatmap (corr-matrix, annot = True, cmap = 'coolwarm',
        fmt = ".2f")
plt.title ("correlation Heatmap")
plt.show()
```

Result:

Thus, the basic preprocessing and EDA on titanic dataset using Pandas, Seaborn, Sklearn tools.